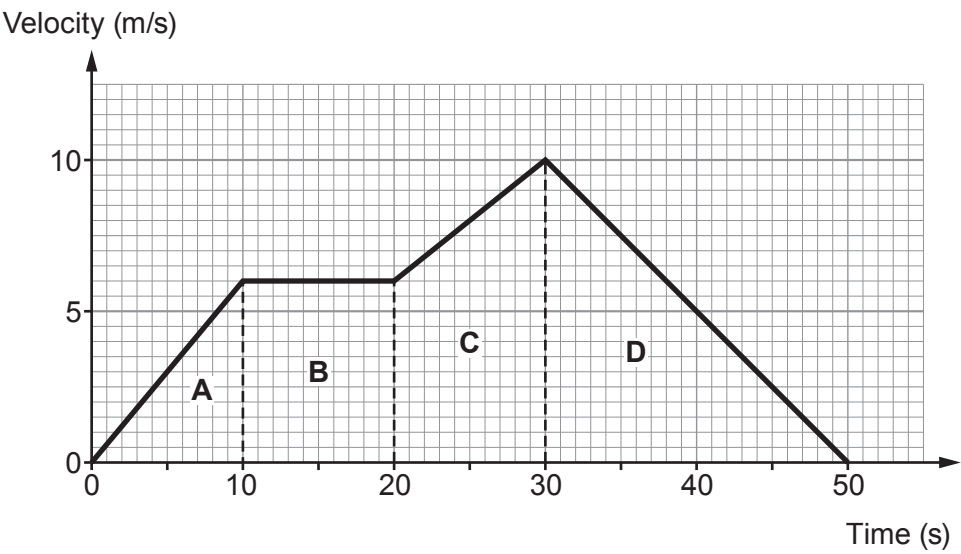


2. The velocity-time graph is for part of a bus journey.



Use the information in the graph to answer the following questions.

- (a) **Complete the table** by placing **one** tick (✓) in each row to describe the motion in each region of the graph. Region A has been completed as an example. [3]

Region of graph	Not moving	Constant velocity	Accelerating	Decelerating
A			✓	
B				
C				
D				



(b) **Complete the following sentences** using numbers from the box.

[3]

2	4	6	8	10	20	50
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(i) The maximum velocity of the bus is m/s.

(ii) The change in velocity of the bus in region C is m/s.

(iii) The bus accelerates for a **total time** of s.

(c) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration in **region A**.

[3]

$$\text{acceleration} = \dots\dots\dots \text{m/s}^2$$

(d) The bus travelled 270 m in the 50 s shown.

Use the equation:

$$\text{mean speed} = \frac{\text{distance}}{\text{time}}$$

to calculate the mean speed of the bus.

[2]

$$\text{mean speed} = \dots\dots\dots \text{m/s}$$

